

Functional Logic – Numerical Operators & Domain Mapping

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27.02.2026

Part I – Numerical Operators (Universal Module)

Normalized Stability Index

We define a normalized stability index:

$$\Phi^*(t) = w_G G^*(t) + w_R R^*(t) + w_M M^*(t), \quad w_G + w_R + w_M = 1,$$

$$G^*, R^*, M^* \in [0, 1].$$

Proxy Normalization

Each proxy component is normalized as:

$$X^* = \text{clip} \left(\frac{X - \mu_X}{\sigma_X} \cdot s_X + b_X, 0, 1 \right),$$

where:

- μ_X, σ_X : reference mean and standard deviation
- s_X : scaling factor
- b_X : bias term
- $\text{clip}(\cdot, 0, 1)$ restricts values to $[0, 1]$

Dynamic Stability Corridor and $\Delta\Phi^*$ Criterion

The stability corridor is dynamic and describes the relationship between freedom and order (typically approx. 60–80% freedom and 40–20% order, interpreted as a calibration range rather than a fixed constant).

For two time points $t_1 < t_2$:

$$\Delta\Phi^* = \Phi^*(t_2) - \Phi^*(t_1).$$

A trajectory or intervention is considered stability-enhancing if:

- $\Delta\Phi^*$ matches the predicted direction and magnitude
- the triadic coupling (Generator / Regulator / Mediator) follows the expected sequence, timing, and scale

Universal Operator and Parameter List

Symbol	Meaning	Scope
w_G, w_R, w_M	Triad weights (Generator / Regulator / Mediator)	universal
s_X, b_X	Proxy scaling and bias	project-specific
μ_X, σ_X	Reference mean / variance	dataset-specific
k_2	Love number (tidal deformation)	domain-specific
Q	Quality factor (tidal damping)	domain-specific
$ B $	Magnetic field strength	domain-specific
LOD	Length-of-day drift	domain-specific
e, a	Eccentricity / semi-major axis	domain-specific
U	Uncertainty operator (ensemble / posterior)	universal
I	Inversion operator (Bayesian / hierarchical)	universal
J	Joint retrieval (spectral / polarimetric)	domain-specific

Quality Assurance (Universal)

- Calibration: determine $\mu_X, \sigma_X, s_X, b_X$ using training windows; monitor drift via rolling statistics.
- Uncertainty: apply U via bootstrap or ensemble; report confidence intervals for $\Delta\Phi^*$.
- Audit / versioning: document all changes to weights and normalization parameters with timestamp, hash, and reviewer.

Part II – Domain Mapping Examples

The universal triadic structure (Generator, Regulator, Mediator) remains invariant. Only proxies, units, and data sources differ by domain.

Domain: Cosmology / Planetology

Role	Proxy	Data Source
Generator	Infrared heat flux	Orbital / lander mapping
Regulator	Q, k_2 (tidal parameters)	inversion models
Mediator	Retrieval coherence	joint spectral retrieval

Modified field equation:

$$G_{\mu\nu} = \kappa T_{\mu\nu} + \varepsilon F_{\mu\nu}(x, B), \quad |\varepsilon| \ll 1.$$

Falsifiable prediction:

$$f\sigma_8(z) = \Omega_m(z)^{0.55} (1 + \varepsilon\Psi(z)).$$

Domain: Human Medicine / Cell Biology

Role	Proxy	Measurement
Generator	ATP production	respirometry
Regulator	Membrane potential / homeostasis	ion markers
Mediator	DNA / signaling integrity	NGS / proteomics

Domain: AI / Machine Learning

Role	Proxy	Measurement
Generator	Compute energy / activation density	run statistics
Regulator	Regularization / guardrails	constraint parameters
Mediator	Alignment / coherence score	evaluation sets

Domain: Sociology / Economics

Role	Proxy	Measurement
Generator	Innovation / capital flow	time series
Regulator	Regulatory intensity	policy index
Mediator	Information network consistency	consensus metrics

Domain: Ecology / Climate

Role	Proxy	Measurement
Generator	Primary productivity	satellite / flux
Regulator	Feedback / damping (albedo)	satellite / in-situ
Mediator	Biodiversity index	survey / posterior

Decision Form (Universal)

$$\Delta\Phi^* = \Phi^*(t_2) - \Phi^*(t_1)$$

A decision is confirmed if:

- predicted direction matches observed direction

- confidence intervals overlap expected range
- triadic coupling sequence is consistent

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